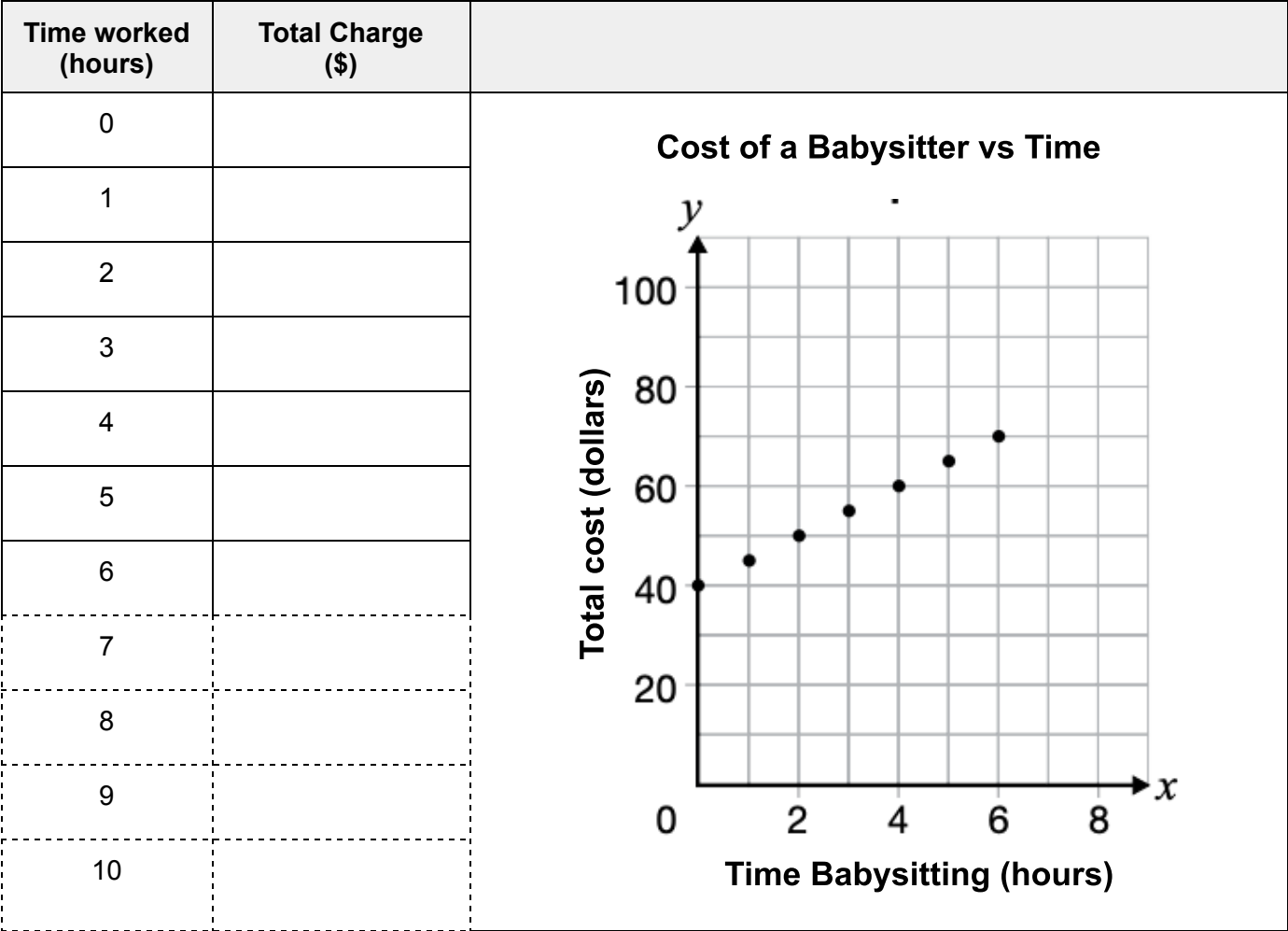


Reading Linear Graphs

The following is a graph of the price of hiring a Babysitter.

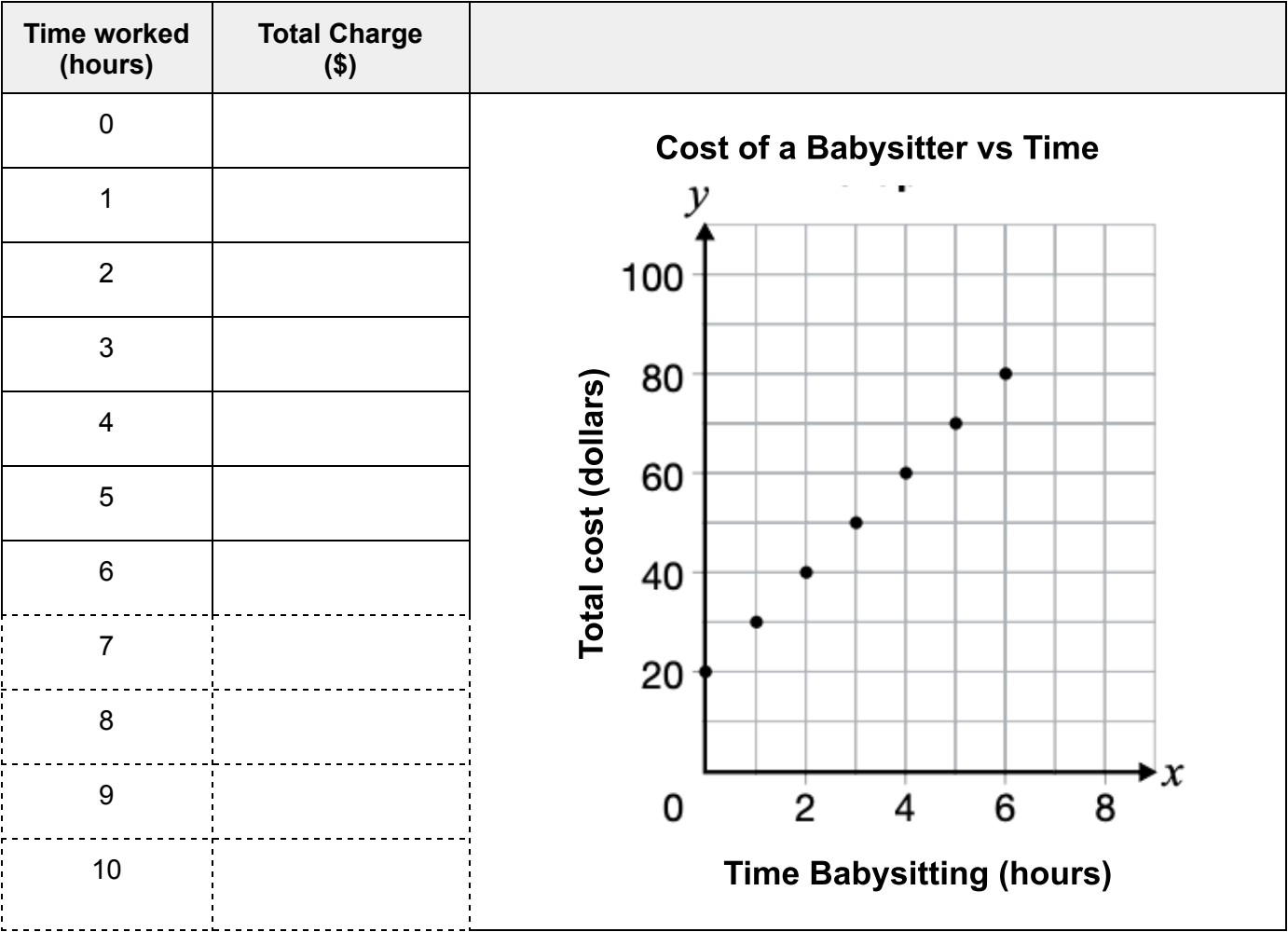
- a) The first **seven rows** can be read from the graph.
Then, **extrapolate** to calculate the cost for more hours than the graph shows.



- b) What is the **fixed cost** (initial cost, the amount you pay at 0 hours) ?
- c) What is the **variable cost** (the amount of money PER hour) ?
- d) The total cost y after x hours of babysitting is: $y = \quad x$

Practice

1. The following is a graph of the price of hiring a Babysitter.
- a) The first **seven rows** can be read from the graph.
Then, **extrapolate** to calculate the cost for more hours than the graph shows.



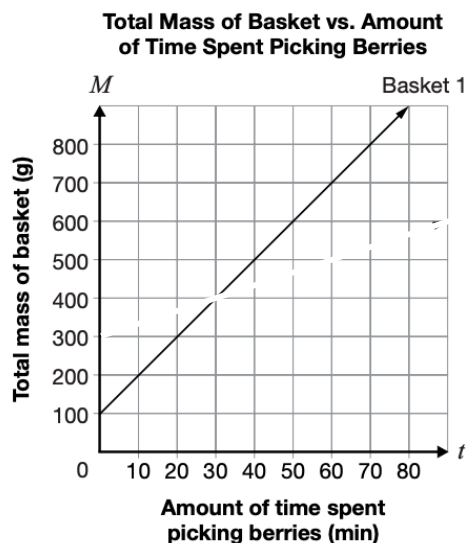
- b) What is the **fixed cost** (initial cost, the amount you pay at 0 hours) ?
- c) What is the **variable cost** (the amount of money PER hour) ?
- d) The total cost y after x hours of babysitting is: $y =$ x

2. This graph shows the relationship between the total mass of a basket, in grams, and the amount of time spent picking berries, in minutes, for two baskets.

a) How much does Basket 1 weigh empty?

b) How many grams of berries are picked every 10 minutes?

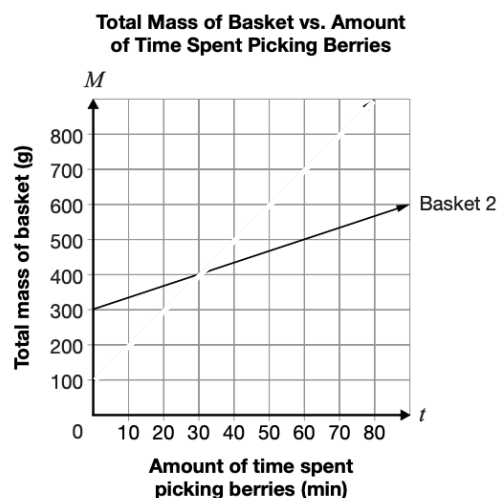
c) How many grams of berries are picked every 1 minute?



3.a) How much does Basket 2 weigh empty?

b) How many grams of berries are picked every 10 minutes?

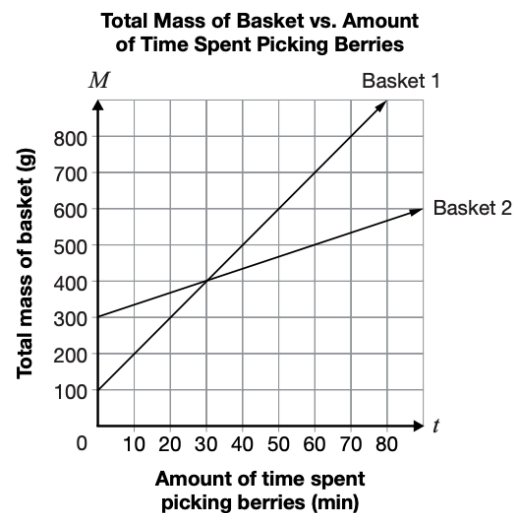
c) How many grams of berries are picked every 1 minute?



4. Here are both graphs on the same axes.

a) At what **amount of time** do Basket 1 and Basket 2 weigh the same?

b) **How many total grams** are the baskets when they are equal?



5. This table of values shows a linear relationship between the cost of an activity and the number of people participating.

Number of people	Cost of activity (\$)
0	50
10	250
20	450

- a) What is the **cost per person** of this activity?
- b) What is the **fixed cost** of this activity?
- c) Create an equation for the total cost (y) of x people doing the activity.
- d) How much would it cost for 40 people to do this activity?

6. This table of values shows the cost (C) of hiring someone for h hours.

Hours Worked (h)	Total Cost (C)
2	\$110
3	\$145
6	\$250
8	\$320

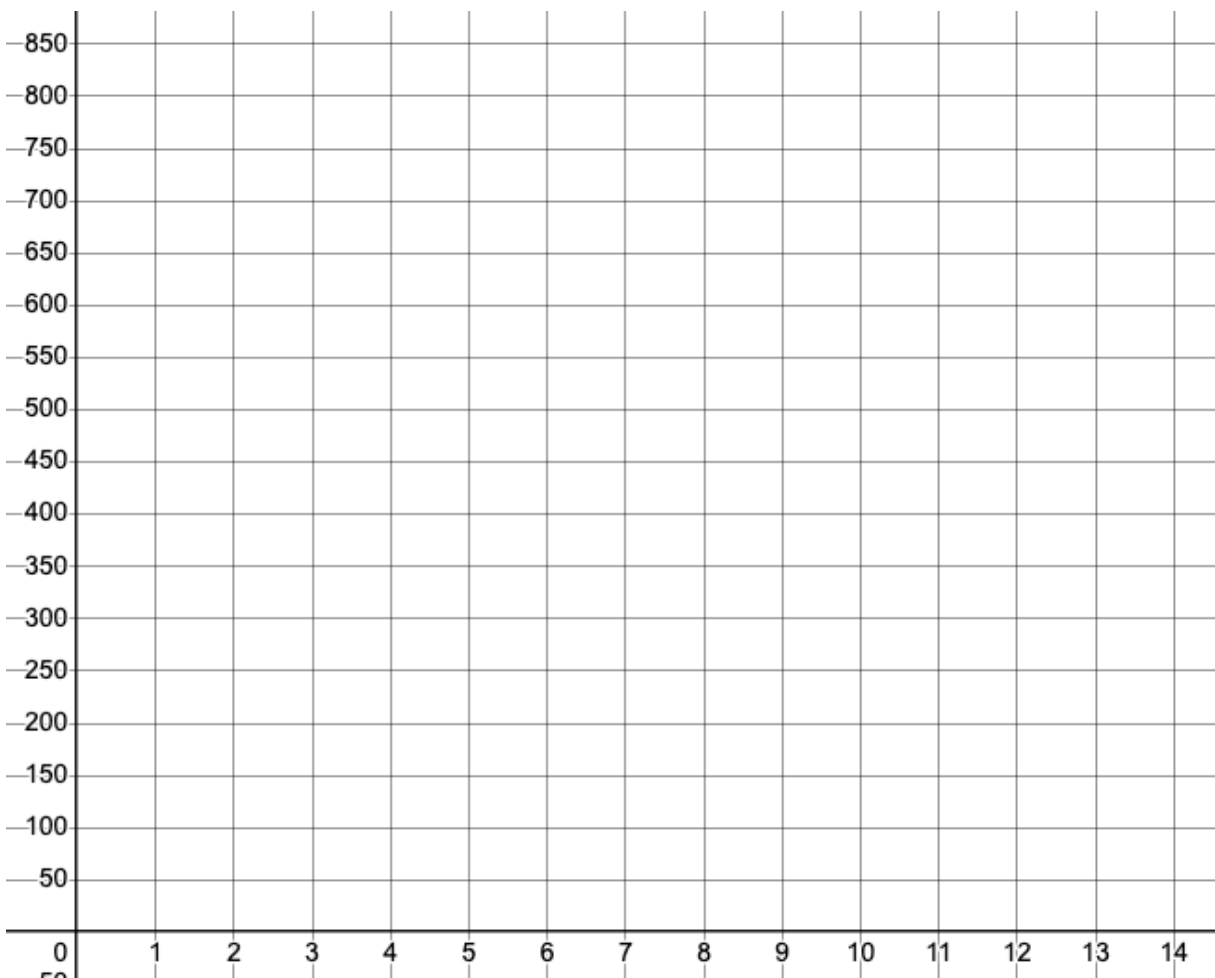
- a) What is the **cost per hour** for this person?
- b) What is the **fixed cost** for this person?
- c) Create an equation for the total cost (C) of hiring this person for h hours.
- d) How much would it cost to hire this person for 16 hours?

7. An event planner offers the following two pricing options:
- Option A: A one-time up-front payment of \$200 *plus* \$30 for every hour worked.
 - Option B: No up-front payment, then \$55 for every hour worked.
- a) Create a graph to compare the total cost of hiring the event planner using **each** of the two pricing options from 0 hours to 14 hours.

Give your graph a title

Label the x- and y-axes

Label the two lines (Option A and Option B)



- b) If an event is expected to require 10 hours of planning, which option will result in a lower cost to the customer, and by how much?
- c) Explain the meaning of the point where the two lines intersect.

8. The cost to print a book on printer A includes a set-up cost of \$225 plus \$6 per page.
The cost to print the same book on printer B includes a set-up cost of \$375 plus \$5.50 per page.
- a) How much does it cost to print a book of each length, with each printer?

Number of Pages	Total Charge with Printer A (\$)	Total Charge with Printer B (\$)
0		
100		
200		
300		
400		
500		
600		

Graph this data; titles; labels, the usual.

